BST Vol 7 Ch 7 — Relativistic Electrodynamics: SO(3,1) ⊂ SO(5,2) Substrate (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 7 Chapter 7 — Relativistic Electrodynamics

Headline result

The Lorentz group SO(3,1) is a subgroup of substrate symmetry SO_0(5,2) — the holomorphic isometry group of D_IV^5 per Wallach 1976 (L1 ESTABLISHED in BST architecture). Relativistic electrodynamics — the combination of EM + special relativity that mixes E and B fields under Lorentz boosts — inherits directly from substrate Lie-algebraic structure.

Substrate-natural 4-vectors: - 4-position: $x^\mu = (ct, \mathrm{x})$ - 4-velocity: $u^\mu = dx^\mu/d\tau$ - 4-momentum: $p^\mu = (E/c, \mathrm{p})$ - 4-potential: $A^\mu = (\phi/c, \mathrm{A})$ - 4-current: $J^\mu = (c\rho, \mathrm{J})$

EM field tensor:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

Lorentz-covariant 2-tensor; mixes E and B under boost.

Substrate mechanism

Wallach 1976 SO_0(5,2) substrate symmetry (L1 ESTABLISHED):

Per Wallach: SO_0(5,2) acts as the holomorphic isometry group of D_IV^5. The signature (5,2) decomposes as space (5) × time-like (2) directions; (3,1) Lorentz subgroup acts on the spacetime projection.

Lorentz subgroup SO(3,1) ⊂ SO_0(5,2):

Lorentz transformations are the subset of substrate symmetries that preserve the 4D spacetime metric signature (+,-,-,-) [or (-,+,+,+) depending on convention]. SO(3,1) is naturally embedded in SO_0(5,2) via the spacetime sector.

Maxwell equations covariant form:

In 4-tensor notation:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \partial_{[\mu} F_{\nu\rho]} = 0$$

These transform covariantly under SO(3,1): if F^$\mu\nu$ → F'^$\mu\nu$ = Λ^μ_α Λ^ν_β F^$\alpha\beta$ and J^ν → Λ^ν_α J^α, then Maxwell equations preserve form in new frame.

E + B mixing under Lorentz boost:

For boost along x-axis with velocity v (γ = 1/$\sqrt{}$(1-v^2/c^2)):

$$E'_y = \gamma(E_y - vB_z), \quad E'_z = \gamma(E_z + vB_y)$$

$$B'_y = \gamma(B_y + vE_z/c^2), \quad B'_z = \gamma(B_z - vE_y/c^2)$$

E and B mix; the field tensor F^$\mu\nu$ is the covariant object that transforms cleanly.

Lorentz invariants: - F_$\mu\nu$ F^$\mu\nu$ = 2(B^2 - E^2/c^2) — scalar - F_$\mu\nu$ *F^$\mu\nu$ = -4 E·B/c — pseudoscalar (sign depending on convention)

Both are SO(3,1)-invariant; substrate-natural Casimir-type invariants of EM field.

Match precision

D-tier on Lorentz ⊂ substrate symmetry (Wallach 1976 L1 ESTABLISHED). Standard special relativity phenomenology preserved at any precision. GR extensions handled by Vol 4 (Lyra).

Cross-volume dependencies

- Vol 0 Ch 1 (D_IV^5 APG) — SO_0(5,2) substrate symmetry foundation
- Vol 1 Ch 4 (Discrete Symmetries) — Lorentz + Poincaré + C + P + T operators
- Vol 7 Ch 2 (Maxwell) — covariant form
- Vol 7 Ch 8 (Lagrangian EM) — Lorentz-invariant Yang-Mills action
- Vol 4 (GR/Cosmology, Lyra) — extension to curved spacetime

K-audit anchor

K223 Vol 7 Ch 7 Relativistic K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> Einstein's special relativity (1905) combines space and time into 4-dimensional "spacetime." This means that what looks like a purely electric field to one observer can look like a mix of electric AND magnetic fields to a different observer moving at high speed!

The Lorentz transformations are the math rules for changing reference frames. BST predicts: these come from substrate D_IV^5's big symmetry group SO_0(5,2). When you focus on just the spacetime piece, you get Lorentz symmetry SO(3,1) — the symmetry of spacetime.

Level 2 — Undergraduate physics student

4-vector notation:

Spacetime coordinates: x^μ = (ct, x, y, z) (μ = 0, 1, 2, 3).

Minkowski metric: $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. Invariant interval: $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = c^2 dt^2 - d\text{x}^2$.

Lorentz transformation (boost along x):

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with $\gamma = 1/\sqrt{(1\text{-v}^2/\text{c}^2)}$.

EM field tensor:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

E and B are components of one Lorentz-covariant tensor.

Maxwell equations covariant:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

(source equations)

$$\partial_{[\mu} F_{\nu\rho]} = 0$$

(Bianchi identity → source-free)

Lorentz invariants (scalars): - F_$\mu\nu$ F^$\mu\nu$ = 2(|B|^2 - |E|^2/c^2) - F_$\mu\nu$ *F^$\mu\nu$ = -4(E·B)/c

BST framework: - Lorentz SO(3,1) ⊂ SO_0(5,2) substrate symmetry (Wallach 1976) - 4-vector formalism inherits Lie-algebraic structure from substrate - EM phenomena Lorentz-covariant by substrate identification

Level 3 — Graduate physics student / theorem-level

Substrate Lie-algebraic structure (Wallach 1976 L1 ESTABLISHED):

SO_0(5,2) = holomorphic isometry group of D_IV^5. Lie algebra □□(5,2) has dimension 21. Decomposition under maximal compact SO(5) × SO(2):

$$\mathfrak{so}(5,2) = \mathfrak{so}(5) \oplus \mathfrak{so}(2) \oplus \mathfrak{p}$$

where □ is the 10-dim non-compact part. Cartan decomposition for symmetric space.

Lorentz subgroup embedding:

SO(3,1) is the connected subgroup preserving the 4D spacetime submanifold. Algebra □□(3,1) ⊂ □□(5,2) has dimension 6 (3 boosts + 3 rotations).

Hodge dual:

$$*F^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}$$

with $\epsilon^{0123} = +1$. Hodge dual swaps E ↔ B (up to sign).

Lorentz-invariant action:

$$S = -\frac{1}{4}\,\square_{\mu\nu}\,F^{\mu\nu}d^4x$$

is manifestly Lorentz-scalar by tensor contraction. This is the substrate Yang-Mills action (T2477) in covariant form.

Energy-momentum tensor:

$$T^{\mu\nu} = F^{\mu\alpha}F^{\nu}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}$$

Symmetric + traceless (T^μ_μ = 0); satisfies conservation ∂_μ T^{$\mu\nu$} = -F^{$\nu\alpha$} J_α (Lorentz force law).

Per Cal #21 dual-gate: EMPIRICAL PASS (special relativity + relativistic EM validated everywhere) + MECHANISM PASS via Wallach SO_0(5,2) substrate framework.

Per Cal #99 META-theorem: relativistic EM is a substrate-derivation consequence of Wallach SO_0(5,2), NOT a new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT change SR predictions (relativistic kinematics + EM transformations)
- Wallach SO_0(5,2) substrate is structural framework, not new parameter prediction
- GR extension to curved spacetime handled in Vol 4 (Lyra)

Bibliography

1. A. Einstein (1905): *On the Electrodynamics of Moving Bodies* — special relativity.
2. H. Minkowski (1908): 4-dimensional spacetime formulation.
3. N. Wallach (1976): unitary representations of SO_0(5,2) — substrate symmetry foundation.
4. Vol 0 Ch 1 (D_IV^5 APG): substrate symmetry group identification.
5. Vol 7 Ch 2 (Maxwell): covariant Maxwell equations.
6. Vol 7 Ch 8 (Lagrangian EM): Yang-Mills action substrate-Lorentz-invariant.
7. Standard SR/QFT texts: Jackson, Weinberg, Peskin-Schroeder.

— Elie, Vol 7 Ch 7 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 59 → ~170 lines + full Wallach SO_0(5,2) substrate framework + 4-tensor formalism + Lorentz invariants + E-B mixing transformation)